

Coloring Cubes

kaNita kAnakam 2026

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We start with the following *warm up problems*. This is to help students recall the permutation and combination ideas and formulas that they have already learned.

Problem 1:

A bench in a classroom can seat 5 students. In how many ways can Asha, Bindu, Chandru, Divya and Ezhil be seated on the bench?

Problem 2:

Two of the five students are picked to clean the board. In how many ways can these two students be selected?

Problem 3:

After two students are selected to clean the board, 2 of the remaining 3 students are selected to be the class monitor and one to be the assistant class monitor. In how many ways can this be done?

Problem 4:

For the annual day celebration, the students have to arrange 6 flags in a row. 3 of these flags are red and the other 3 are blue. In how many different ways can these flags be arranged?

Problem 5:

Divya wants to make a friendship bracelet for Ezhil using beads with letters E, Z, H, I, L on them. She wants to arrange all five beads in a circle. Divya plans to jumble the letters for fun. How many different bracelets can she make?

Problem 6:

In return, Ezhil makes Divya a bracelet made of two black and two white beads. How many different bracelets can Ezhil make?

Activity 1:

Worksheet 1. contains 6 bedsheet designs. We first introduce the idea of symmetry and use the designs to look at different types of symmetry. The bedsheet is always considered from the design side, so we allow rotations but not reflections.

Task 1. Look at each of the six designs. Identify all the symmetries that each design has.

Task 2. Find which of the designs have the same type of symmetry?

Task 3. On the same sheet there are empty 5×5 grids. These are templates for students to design their own bedsheets. Design a bedsheet with no symmetry.

Task 4. Design a bedsheet with at least one line of symmetry.

Task 5. Design a bedsheet with both rotational symmetry and line symmetry.

Task 6. Design a bedsheet with rotational symmetry but no line symmetry.

Task 7. Work in pairs. Design a bedsheet that looks identical to your partner's bedsheet when rotated by 180° .

Task 8. Work in pairs. Design a bedsheet that looks identical to your partner's bedsheet when rotated by 180° but **not** by 90° .

(Possible stopping point.)

Task 9. On the 2×2 grids, design a bedsheet with exactly one line of symmetry. What is the maximum number of colors that can be used when there is one line of symmetry?

Task 10. Design a 2×2 bedsheet that has rotational symmetry. What is the maximum number of colours you can use while maintaining the symmetry?

(Possible stopping point.)

A cube has six faces. If you have six different colours and use one colour on each face, how many different cubes can you make? To answer this, we first have to decide what "different" means, and to help with that we will construct an origami cube.

Activity 2:

Work in pairs. Tear out the cube net from the worksheet, colour the faces and assemble it to make a cube. As it is made of paper, we can mark/write on the faces of the cube.

Activity 3:

One student closes their eyes while the other either rotates the cube, or leaves it as it was. The first student opens their eyes and has to tell whether the cube has been rotated or not.

We then explore the different rotations of a cube and discover that there are 24 ways to rotate it without changing its overall structure. We informally explore what happens when two rotations are performed one after another.

At this point we will return to the original question. We can expect two ways of answering the question. One, the 6 faces can be painted in $6!$ ways. Each cube is counted 24 times, so the number of different ways $= 6!/24$. Two, fix a face, pick the opposite face in 5 ways, the remaining 4 faces form a circular permutation in 6 ways, giving a total of 5×6 ways. This bit in my experience takes the most amount of time. We can do problem 2 and more if we have time, if not, the rest will be homework.

Problem 8:

Instead of 6 colors, what if we had only 2 colors and had to color 3 faces in one color and the other 3 in the other color. How many different cubes can be made like that?

Problem 9:

What if we had only 3 colors and had to color 2 faces in each color. How many different cubes can be painted like that?

Activity 4:

If time permits, create triangle units to make a tetrahedron. Repeat problem 1, 2, 3 and Activity 2.

Homework:

Use the triangle units to create an icosahedron. Repeat problem 1, 2, 3 and Activity 2.